



Term Evaluation Theory (Even) Semester Regular Examination February 2026

Roll no.....

Name of the Course: M. Tech
Semester: II
Name of the Paper: Advanced Operating System
Paper Code: MCS-202
Time: 1.5 hour

Maximum Marks: 50

Note:

- (i) Answer all the questions by choosing any one of the sub-questions
- (ii) Each question carries 10 marks.

- Q1. (10 Marks)
- a. Explain the architecture of a time-sharing operating system with a neat diagram. CO1
- OR
- b. Define a process. Explain the process life cycle with a neat diagram. Describe the role of the Process Control Block (PCB). CO2
- Q2. (10 Marks)
- a. Compare Time Sharing Systems, Batch Systems, and Real-Time Systems. CO1
- OR
- b. Differentiate between processes and threads. Explain user-level threads and kernel-level threads with suitable examples. CO2
- Q3. (10 Marks)
- a. What is the critical section problem? Explain the three necessary conditions for its solution with suitable examples. CO2
- OR
- b. Define semaphores. Explain binary semaphore and counting semaphore operations with suitable pseudo-code. CO2
- Q4. (10 Marks)
- a. Explain the advantages of multithreading in modern operating systems. Illustrate how multithreading improves system responsiveness and resource utilization. CO1
- OR
- b. Explain the concept of Time-Sharing Operating Systems. Illustrate how CPU scheduling enables interactive performance in such systems. CO1
- Q5. (10 Marks)
- a. Explain how advancements in hardware technologies (processors, memory, and storage) have influenced the evolution of operating system strategies. CO1
- OR
- b. Explain the user's perspective of an operating system. Discuss how operating systems simplify user interaction and resource utilization with suitable examples. CO1